EXPERIMENT-9
16-BIT ADDITION

AIM: To write an assembly language program to implement 16-Bit addition using 8086 processor.

ALGORITHM:

1. Start the program by loading a register pair with address of 1st number.
2. Copy the data to another register.
3. load the ~~se~~ second number to the first register.
4. Add the two register pair contents.
5. check for carry.
6. Store the value of sum and carry in memory location. Results store in AX.
7. Terminate the program.

PROGRAM:

```
MOV AX,[1100H] - load first 16-bit number
MOV BX,[1102H] - load second 16-bit number
ADD AX, BX - Add AX and BX
MOV [1200H], AX - Store sum in memory
HLT - Halt program execution.
```

Input:

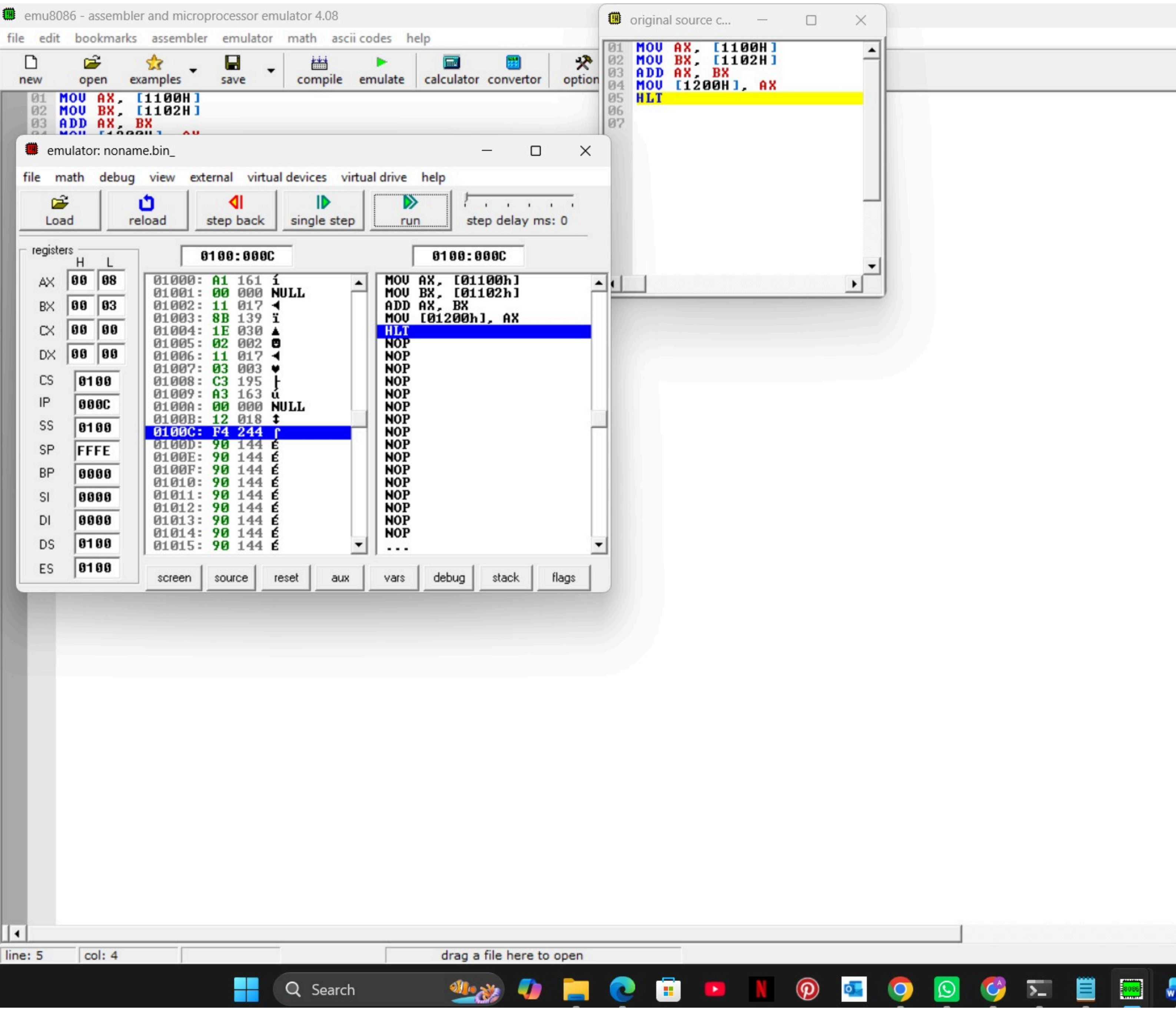
Register	memory	Data
AX	32	1100
BX	45	1102

0117

Output:

Register	memory	Data
AX	77	1200

RESULT: Thus the program was executed successfully using 8086 process simulator.


emu8086 - assembler and microprocessor emulator 4.08
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new open examples save compile emulate calculator convertor option
01 MOV AX, [1100H]
02 MOV BX, [1102H]
03 ADD AX, BX
original source c...
01 MOV AX, [1100H]
02 MOV BX, [1102H]
03 ADD AX, BX
04 MOV [1200H], AX
05 HLT
emulator: noname.bin_
file math debug view external virtual devices virtual drive help
Load reload step back single step run step delay ms: 0
registers
H L
AX 00 08
BX 00 03
CX 00 00
DX 00 00
CS 0100
IP 000C
SS 0100
SP FFFE
BP 0000
SI 0000
DI 0000
DS 0100
ES 0100
0100:000C
0100:000C
01000: A1 161 í
01001: 00 000 NULL
01002: 11 017
01003: 8B 139 ï
01004: 1E 030
01005: 02 002
01006: 11 017
01007: 03 003
01008: C3 195
01009: A3 163 ú
0100A: 00 000 NULL
0100B: 12 018
0100C: F4 244
0100D: 90 144 É
0100E: 90 144 É
0100F: 90 144 É
01010: 90 144 É
01011: 90 144 É
01012: 90 144 É
01013: 90 144 É
01014: 90 144 É
01015: 90 144 É
MOV AX, [01100h]
MOV BX, [01102h]
ADD AX, BX
MOV [01200h], AX
HLT
NOP
...
screen source reset aux vars debug stack flags
line: 5 col: 4 drag a file here to open
Search